Choosing among a site's radiocarbon dates can create or reverse regional trends

Abstract

Regional radiocarbon analyses reconstruct past human activity from dated evidence, but date counts do not reveal how dates were chosen. We show that choosing among completed determinations can create or reverse an inferred trend. In simulations of constant and changing histories, retaining every site's oldest date imposed decline and retaining its youngest imposed growth in every replicate, even when models used full calibrated age distributions. Partial retention could create apparent change or reverse weak trends. At half-site treatment, the same rules created apparent change from constant histories in over 97% of replicates; with three-quarters of sites treated, keeping the extreme opposed to the trend reversed 93–100% of 1.5- and 2-fold changes, though no ten-fold change. Random retention reversed at most 0.6% of replicates in any changing-history scenario. Keeping both extrema avoided reversals but obscured weak changes, and at full treatment exponential models were favoured for every ten-fold linear history. In 540 charcoal or wood dates from 68 Australian sites, oldest and youngest retention shifted the mean of the summed probability distribution 3,133 years older and 2,439 years younger; one random date per site shifted it about 100 years. Full-site oldest and youngest retention changed temporal shape more than the choice among three summary methods in all nine sensitivity configurations. The simulations isolate independent regional dates without occupation histories; their ten-fold headline results held under realised measurement scatter. Regional inference needs the selection history of its inputs: equal date counts and agreement among methods do not establish temporal representativeness.

Keywords

radiocarbon dating; palaeodemography; dataset formation; date retention; order statistics; model selection; simulation.

1. Introduction

From the same 540 radiocarbon determinations at 68 Australian archaeological sites, keeping each site's oldest date or its youngest produces summed probability distributions whose means lie more than 5,500 years apart, although each subset contains exactly 68 dates. Keeping one date per site at random shifts the mean by about a century from the full-set baseline, averaged over repeated retention. A retained age alone does not reveal the rule that kept it, yet curves built from such dates are read as histories of human activity. Controlling how many dates a site contributes does not control which ages it contributes.

"Dates-as-data" inference uses the frequency of radiocarbon determinations to investigate human activity and, by proxy, population size: other things being equal, more people leave more dateable evidence (Rick, 1987; Robinson et al., 2019; Ulm and Hall, 1996; Williams and Ulm, 2016). Applications include European Neolithic demography (Shennan et al., 2013; Timpson et al., 2014), Holocene Sahul (Johnson and Brook, 2011; Williams, 2013, 2012), other regional histories (Brown and Crema, 2019; Chaput and Gajewski, 2016; Porcic et al., 2016) and continental syntheses (Wirtz et al., 2024). Each determination is calibrated to a probability distribution over calendar years, often with several peaks because atmospheric radiocarbon has varied through time. Regional summaries include summed probability distributions (SPDs), which add these distributions, frequency distributions (FDs) and kernel density estimates (KDEs); explicit population and occupancy models offer further approaches (Bronk Ramsey, 2017; Brown, 2015; Crema, 2022; Mendez-Quiros et al., 2023; Michczynska et al., 2007).

The connection between dated evidence and people depends on preservation, research effort and analytical choices. Younger deposits are more likely to survive, producing a "pull of the recent" (Bluhm and Surovell, 2019; Surovell et al., 2009). Unequal sampling among settlements, calibration, binning and model choice also affect inference (Carleton and Groucutt, 2021; Park et al., 2026). Taphonomic adjustments, simulation tests and count-based models address aspects of these problems (Crema, 2025; De Navascués et al., 2025; Heaton et al., 2025; Peros et al., 2010; Williams et al., 2015). Each analysis nevertheless inherits decisions about which completed determinations enter its dataset.

Date counts alone cannot reveal those decisions. Counts vary with budgets, preservation, research priorities and the contexts chosen for dating (Codding et al., 2024). In the original AustArch compilation, 53% of sites had one determination and 86% had fewer than four (Williams et al., 2014). A single record may be the only date obtained, or one retained from a larger set. Rescaling contributions for unequal sampling intensity (Park et al., 2026) addresses how much a site contributes, while age-dependent retention

changes its temporal contribution.

The underlying statistical mechanism is established: an extreme observation is distributed differently from an arbitrary member of a sample (Fig. 1). Of five independent ages from a constant distribution, the oldest falls in the oldest fifth of the range about two times in three, against one in five for a random date. Related issues arise when material near layer boundaries is chosen for chronological modelling (Nicholls and Jones, 2001) and when last occurrences are used to estimate extinction times (Bradshaw et al., 2012; Signor and Lipps, 1982). Our experiment concerns selection among completed determinations. It differs from choosing a basal context before dating, and an extreme calibrated date need not mark occupation onset or termination.

We test how this retention process propagates through calibration and regional inference, how its effect depends on the strength of the true trend, and whether keeping both ends of a site's dated sequence avoids the problem. Simulations provide known constant, increasing and decreasing histories; an AustArch benchmark (Williams and Ulm, 2014) measures deformation in an assembled archive. We show that full-site oldest or youngest retention can determine the inferred direction, even when models use full calibrated age distributions, and that retention at only some sites can reverse weak trends. Keeping both extrema avoids reversal but obscures weak trends, and for ten-fold changes it makes linear histories favour exponential models. In the empirical benchmark, oldest and youngest retention moved regional curves further than the choice among three summary methods. Agreement among analyses of the same dates cannot, by itself, show that those dates represent the history being inferred.

2. Methods

The simulations test inference against known histories; the empirical comparison measures deformation of the same archive. Figure 1 illustrates exact-age selection with five dates per site (Supplementary Section S2.6); it is not a calibrated simulation result. Detailed filters and reproducibility records are in the Supplementary Information.

2.1 Empirical benchmark

We began with a 976-determination, 116-site Holocene reference set derived from the Archaeology Data Service AustArch file `Austarch_1-3_and_IDASQ_28Nov13-1.csv`; the derivation and its provenance are documented in the Supplementary Information and repository (Williams et al., 2014; Williams and Ulm, 2014). For the primary empirical benchmark, we retained terrestrial charcoal or wood records coded as occupation-related (`Y`) or of uncertain occupation relation (`U`), excluding shell and human bone. All retained dates were calibrated with the Southern Hemisphere curve SHCal20 (Hogg et al., 2020). We then required a median of the full calibrated posterior within 751–11,550 calendar years before present (cal BP; present = AD 1950), at least 0.80 posterior mass within the 651–11,650 cal BP analysis window, and at least five eligible determinations per site. The common-membership benchmark contained 540 dates from 68 sites (181 `Y`, 359 `U`), with identical dates entering every summary method. Restriction to well-replicated sites precludes treating it as an unbiased sample of Sahul occupation.

We constructed three unit-integral summaries on common 100-year bins. The SPD summed each calibrated posterior within the analysis window and applied a centred 100-year running mean, the FD binned posterior medians with a two-bin running mean for display, and the KDE placed Gaussian kernels of 100-year bandwidth on the same medians, reflected at both window boundaries to limit edge bias. Every method received the same determinations in every realisation. The curves describe relative temporal shape, because normalisation removes information about the number of retained dates. They serve only to describe the empirical benchmark; in the simulations, models are fitted to individual calibrated age distributions, not to the SPD, FD or KDE curves.

2.2 Empirical retention experiment

We compared all dates with five post hoc retention rules: random-1 and random-5 (one or five dates chosen at random from each treated site), oldest, youngest and bracketed (oldest plus youngest). A rule was applied to 50% or 100% of sites; untreated sites retained all eligible dates. Random retention and random selection of half the sites used 1,000 deterministic-seed realisations, whereas full-site oldest, youngest and bracketed scenarios were deterministic. Extrema were defined by the median of the full calibrated posterior, with posterior mean and then stable record identifier breaking ties. A median summarises a posterior that may be multimodal, so "oldest" and "youngest" are rules about calibrated summaries, not claims about the true age order within a site.

Figure 6 plots retained-minus-baseline curves; its caption defines the conditional retention envelopes. We quantified curve separation using total variation,

$$D_{\mathrm{TV}}(p, q) = \frac{1}{2} \sum_{t} |p_t - q_t|.$$

Here p_t and q_t are the normalised probability masses in bin t, and D_{TV} is the share of mass that must be moved to turn one curve into the other: zero for identical curves and one for non-overlapping curves. For each realisation, a conservative contrast C subtracted the largest between-method distance on the retained dates from the smallest method-specific retention distance to its all-date baseline. Retention was judged to matter more than the choice of method only when C was positive (for stochastic scenarios, when its 2.5th percentile was positive): that is, when even the least-affected method had moved further from its own baseline than any two methods differed on the same retained dates. Mean-cal-BP shifts measured temporal displacement; positive values are older. Prespecified sensitivities varied the calendar buffer, posterior-mass threshold, KDE bandwidth and boundary correction, and restricted the benchmark to `Y` records.

2.3 Simulation of constant and changing histories

The simulation used the full 976-date, 116-site reference set, which also includes the shell and human-bone determinations excluded from the benchmark, to estimate a truncated negative-binomial distribution of dates per site and a regression of logged reported error on radiocarbon age. Each replicate contained 116 site labels with at least five dates per site. Calendar ages were drawn independently from one of five regional event-age densities over 651–11,650 cal BP, each describing the relative temporal distribution of dated events and integrating to one: uniform, linear increase, linear decrease, exponential increase or exponential decrease (Supplementary Fig. S1). In the main experiment, each non-uniform truth had a ten-fold end-to-end ratio. We summarised every trajectory by the log ratio of density at the recent and old ends of the window,

$$\theta = \log\left\{ \frac{f(651)}{f(11650)} \right\}.$$

Here "increase" means greater density towards the present, at the low-cal-BP end, so ten-fold truths had $\theta = \pm \log(10)$. Sites were labels only: they determined the number of dates and subsequent retention but imposed no occupation windows, stratigraphy or within-site dependence.

The calibration-curve-mean observation model mapped each calendar age to the SHCal20 expected conventional age, simulated a reported one-sigma error from the fitted age–error relation, and recalibrated it. Each simulated determination therefore carried a quoted error drawn from the empirical model and the full calibrated-posterior shape, but its central conventional age was the curve mean at the true calendar age, with no added measurement scatter. An earlier paired sensitivity added realised scatter only for constant and increasing histories and the all-dates and random rules (Supplementary Section S3); a later paired rerun applied it to the whole main experiment (Supplementary Section S2.8).

We compared seven retention rules with an all-dates reference: random-1, random-2, random-3, random-5, oldest, youngest and bracketed, with extrema selected by posterior median. Each rule treated 50%, 75% or 100% of sites, and untreated sites retained all dates. We generated 1,000 replicates for every truth-by-proportion combination, with common random numbers applying every rule to the same simulated dates within a replicate. A signal-strength extension repeated the four changing histories at end-to-end ratios of 1.5, 2, 3 and 5, with 500 replicates per combination and otherwise identical code, data and seeds, so each replicate shared its latent draws across ratios (Supplementary Section S2.7).

We fitted three functional families (uniform, linear and exponential) directly to the retained calibrated posteriors. Both non-uniform families allowed increase or decrease. For date i, the primary marginal log likelihood was the log of the candidate density averaged over the date's calibrated age distribution,

$$\ell_i = \log \int f(x \mid \theta) p_i(x)\, dx,$$

where $p_i(x)$ is the calibrated posterior. Calibration used a uniform calendar-age prior; posteriors were restricted to the strict interior of the analysis window and renormalised. Conditional on posterior mass within that window, ℓ_i is the observed-data marginal log likelihood up to a date-specific additive constant

independent of θ; these constants cancel in within-dataset AIC comparisons (Timpson et al., 2021). Minimum Akaike Information Criterion (AIC) selected the family (Akaike, 1974). Because AIC penalises each fitted parameter, a linear or exponential family was preferred to the uniform only when its improvement in fit outweighed that penalty.

This criterion treats retained dates as independent regional draws and does not model their retention probabilities. Under bracketed retention, the two extrema retained from a site are dependent, but the product criterion treats them as if they were independent (working independence); bracketed AIC and family-selection results therefore describe the analysed workflow rather than likelihood support under a joint paired-order-statistic model. A separate ℓ^2 analysis used the posterior expectation of the model log density; we call this an expected-log-density scoring sensitivity, not an ordinary likelihood analysis.

We measured correct-direction selection (uniform for a constant history, otherwise the correct direction), selection of the generating family, and the fitted generating-family endpoint contrast. For Figure 4, we used the bounded scale

$$c = \frac{f(651) - f(11650)}{f(651) + f(11650)} = \tanh(\theta/2).$$

This contrast ranges from -1 to 1, remains defined when one endpoint density is zero, and equals $\pm 9/11$ for the ten-fold truths. It describes relative endpoint density, not a growth rate or recovery of the full density shape. Each fit was transformed before calculating means and central 95% replicate ranges; every group shows its near-boundary fraction (Supplementary Section S2.4). The largest possible Monte Carlo standard error for a selection rate is 0.016 with 1,000 replicates and 0.022 with 500.

2.4 Reproducibility

Analyses used R 4.5.3, rcarbon and a pinned package environment (Crema and Bevan, 2021; R Core Team, 2025). Deterministic seeds enable repeatability; checkpoints, source fingerprints and checksum manifests link accepted evidence to its generating code. The review archive provides code, aggregate empirical evidence and synthetic simulation outputs; source archaeological records are governed separately (Data availability).

3. Results

3.1 The retention rule controlled the inferred direction

The simulations show how retention alone can create an apparent regional trend (Fig. 2). From a constant generating history, AIC favoured exponential decline in all 1,000 replicates after retaining every site's oldest date, and exponential growth in all 1,000 after retaining its youngest. The same rules always imposed decline and growth, respectively, on the four changing histories, so a rule opposing the generating trend reversed it in every replicate.

For ten-fold changes, complete reversal required every site to be treated: at 50% and 75% treatment, the opposing rule never reversed a changing history, although it increasingly selected a constant one instead. Constant histories were more sensitive to partial treatment: oldest and youngest retention selected a directional model in 97.3% and 97.9% of replicates at 50%, and their respective exponential directions in every replicate from 75%. All four random rules preserved the direction of every ten-fold change at every treatment proportion. Bracketed retention also preserved those directions and selected uniform in 999 of 1,000 constant-history replicates when every site was treated.

Weaker trends were reversed with fewer sites treated (Fig. 3). Under full treatment, the opposing rule reversed every replicate at every tested ratio from 1.5 to 10. With three-quarters treated, it reversed 93.4–100% of 1.5- and 2-fold changes, 35.2–50.0% of 3-fold changes, and no ten-fold change. At half treatment, it still reversed 19.0–22.6% of 1.5-fold changes. Random retention reversed at most 0.6% of replicates in any changing-history scenario. Bracketed retention caused no reversals, but at full treatment it detected only 10.6–16.8% of 1.5-fold changes, compared with 60.6–66.6% for count-matched random-2. Avoiding reversal therefore did not ensure recovery of a trend. Supplementary Section S2.7 gives the complete results.

Fitted endpoint contrasts show the distortion in relative density (Fig. 4). With all dates and random-5, mean generating-family contrasts remained close to the true $|c| = 9/11 \approx 0.818$. When every site was treated, an opposing oldest or youngest rule reversed the fitted sign, and a rule aligned with the generating

trend drove contrasts towards an endpoint. Bracketed retention attenuated the mean absolute contrast to 0.345–0.351 under linear histories and 0.585–0.597 under exponential histories.

3.2 Correct direction does not ensure correct shape

For ten-fold changes, keeping both extrema preserved direction but changed the favoured form for linear histories (Fig. 5). With this rule applied to every site, AIC favoured exponential models in every replicate generated from either linear or exponential histories: no linear history was recovered as linear. With three-quarters or more of sites treated, none was recovered as linear at any tested signal strength. In the main experiment, the primary marginal-likelihood and expected-log-density analyses agreed on the selected family and direction in 99.75% of 120,000 paired selections, and on the selected direction in every all-site oldest and youngest replicate.

Across the other rules, recovering direction was also easier than distinguishing linear from exponential change. With all dates, generating-family selection rates were 0.974–0.976 for ten-fold linear histories, 0.996–0.998 for ten-fold exponential histories and 0.848 for constant histories. Across the ten-fold changing histories and treatment proportions, random retention preserved direction while generating-family selection ranged from 0.754 to 0.995 as dates were discarded.

3.3 Oldest and youngest retention moved the empirical curves by millennia

Retaining one date per site had very different consequences depending on how that date was chosen (Fig. 6). Keeping each site's oldest date shifted the SPD mean 3,133 years older, and keeping the youngest shifted it 2,439 years younger. Keeping one random date per site shifted the mean by +103 years on average across 1,000 realisations, with a central 95% range of −433 to +594 years. All three rules retained 68 dates, and their SPD total-variation distances from the 540-date baseline were 0.455 for oldest and 0.524 for youngest, compared with a random-1 mean of 0.204 (95% range 0.158–0.256). The count-matched comparisons isolate age selection; ranges describe repeated retention from this fixed benchmark.

All-date distances were 0.046 for SPD–KDE, 0.088 for FD–KDE and 0.101 for SPD–FD (Supplementary Fig. S3). When oldest or youngest retention was applied to every site, its deformation exceeded between-method differences under the prespecified conservative contrast: $C = 0.193$ and 0.370, respectively. This ordering and the direction of the mean shift held in all nine sensitivity configurations, including the 154-date, 24-site occupation-related (`Y`-only) subset.

Treating half the sites produced smaller changes. Mean distances across SPD, FD and KDE were 0.092–0.120 for oldest retention and 0.101–0.130 for youngest, with SPD mean shifts of +349 years (95% range −22 to +738) and −276 years (−673 to +137), respectively. Neither half-site rule met the retention-versus-method criterion, and nor did random retention at either treatment proportion or half-site bracketed retention.

Retaining both extrema from every site gave a smaller temporal displacement: the SPD mean shifted +352 years, within the random-1 shift range, while its distance was 0.279, above random-1's 97.5th percentile of 0.256. Bracketed retention kept 136 dates, twice as many as random-1, so random-1 is not a count-matched test for this rule, and the empirical experiment has no random-2 comparator. Across methods, bracketed distances were 0.279–0.326, and the conservative contrast was positive in eight of the nine configurations, including the primary analysis ($C = 0.084$), but negative in the `Y`-only sensitivity ($C = -0.054$).

4. Discussion

4.1 What agreement among analyses can miss

The full-site oldest and youngest rules imposed their respective directions in every replicate, including when those directions opposed the generating history. Repetition therefore did not remove the error. The fitted likelihood carried uncertainty about each retained date through calibration, but did not represent why that date had been retained. These are different sources of uncertainty: a precise age distribution can belong to an unrepresentative subset. In AustArch, agreement among summaries was likewise insufficient reassurance. Under full oldest or youngest retention, the largest difference between methods was smaller than the smallest change from a method's own all-date baseline.

Keeping both extrema reveals why even an apparently correct answer needs interpretation. For ten-fold changes, the fitted direction was correct in every replicate, but all linear histories favoured exponential

models under full-site retention, and most 1.5-fold changes were read as constant. For exactly observed uniform ages, pooling both extrema from sites with at least three dates produces a U-shaped density outside the fitted candidate set (Fig. 1; Supplementary Section S2.6). Uniform selection in 999 of 1,000 calibrated replicates therefore does not establish shape recovery. This is a finding about the tested candidate set and its working-independence fit; richer candidates, goodness-of-fit diagnostics or a model of retention could reveal or address features missed here.

The order-statistic mechanism itself is long established, and extinction studies already treat a species' last dated occurrence as a biased estimate of its endpoint for this reason (Bradshaw et al., 2012; Signor and Lipps, 1982). Dates-as-data analysis can import the same extremes while treating them as typical. What the experiments add is the consequence for regional inference after calibration: retention can fix the inferred direction, reverse weak trends when only some sites are affected, leave direction intact while changing the favoured form, and move regional curves by millennia in an assembled archive. Model comparison can therefore favour a history created by retention.

4.2 Making selection history part of regional inference

Dataset formation belongs alongside taphonomy, binning, rescaling and population modelling in the interpretation of dates-as-data (Bronk Ramsey, 2017; Carleton and Groucutt, 2021; Crema, 2022; Riris, 2018; Surovell and Brantingham, 2007; Williams, 2012). Count-based and Bayesian approaches offer ways to analyse dated evidence (Arthur et al., 2023; Crema, 2025; De Navascués et al., 2025; Heaton et al., 2025), but evaluating their inputs also requires knowing which dates were available and which were kept. An oldest-date summary assembled to investigate arrival, for example, targets a different quantity from the temporal distribution of dated activity. Reusing it for regional inference carries its original retention rule into the new question.

Archives should distinguish sites dated once from those represented by one selected date, recording whether age influenced retention. Material, period, region and selection history help interpret sparsely dated sites without excluding them solely for having few dates (Bird et al., 2022; Loftus et al., 2019; Lucarini et al., 2020; Saktura et al., 2023). With fuller date sets, count-matched comparisons separate discarding dates from favouring particular ages. The simulations provide these controls for oldest and youngest versus random-1, and for both extrema versus random-2. Where the selection history is unknown, testing plausible retention rules can identify conclusions that depend on it; it cannot recover the missing history or certify an unbiased reconstruction.

4.3 Scope of the experiment

The strength of the underlying trend and the extent of retention interact. Under one-sided retention at partial treatment, each filtered site contributed one date and each untreated site at least five, so filtered sites supplied a smaller share of retained dates than of sites. Untreated sites reduced reversal risk, but the outcome depended strongly on signal strength: at 75% treatment, the opposing rule reversed almost all 1.5- and 2-fold changes but no ten-fold change, and still read up to 21.2% of ten-fold changes as constant. For a fixed proportion of sites treated, weaker trends were more vulnerable to reversal or uniform selection, and the tested proportions do not establish a safe retention threshold. The ratios are end-to-end density contrasts over 11,000 years, not growth rates, and non-monotonic histories were not tested.

The simulations omit occupation durations, stratigraphy, spatial structure and context selection. Their random-retention results establish neither a universal dating minimum nor an optimal allocation of dating effort. The restricted empirical benchmark measures relative temporal shape, not population abundance. Its deformation does not identify the actual population history, and these experiments do not estimate the prevalence of extremal retention.

The main observation model omits realised conventional-age variation and correlated calibration-curve errors. An earlier realised-noise sensitivity, limited to all-date and random rules and growth-only candidates, met 15 of 18 prespecified criteria and remains failed; a post-unblinding diagnostic met revised criteria (Supplementary Section S3). A later prespecified rerun of the main experiment with realised scatter, including oldest, youngest and bracketed retention, moved no headline rate by more than 0.001 (Supplementary Section S2.8). Correlated curve errors, laboratory offsets and outliers remain untested.

4.4 Conclusions

A regional radiocarbon analysis inherits an inferential decision when dates are chosen from each site. In the experiments here, that decision could create a trend, reverse one even when only some sites were affected, or preserve its direction while changing its inferred form. Equal site contributions and

agreement among analytical methods did not resolve the problem. Recording retention history and testing conclusions against plausible alternatives should accompany the modelling of dated evidence, because the meaning of a regional curve depends on how its dates came to represent the past.

Data availability

For peer review, the analysis code, pinned computational environment, synthetic simulation outputs and aggregate empirical evidence are supplied as an anonymised supplementary archive. The archive excludes raw and record-level AustArch data, so rebuilding the empirical curves from source records requires separately governed access to the cited AustArch dataset (Williams and Ulm, 2014). A citable repository record containing the code, synthetic simulation outputs and permitted aggregate evidence will be made public no later than publication. Release of source or record-level archaeological data remains subject to source licensing, sensitive-location review and Indigenous data-governance requirements.

Competing interests

None to declare.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

The authors used OpenAI ChatGPT and Codex (GPT-5.6 Sol, GPT-6 Astra and GPT-6) and Anthropic Claude (Fable 5.1 and Opus 5.5). The systems contributed to analysis design, developed and checked code, executed simulations, produced figures, audited evidence, and drafted and revised the manuscript. The authors provided scientific direction and are responsible for the interpretation, final review and approval. Rerunning the analyses requires the archived code, pinned environment and input data, but no AI tools or original prompts. The authors take full responsibility for the article.

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457 **Fig. 1.**

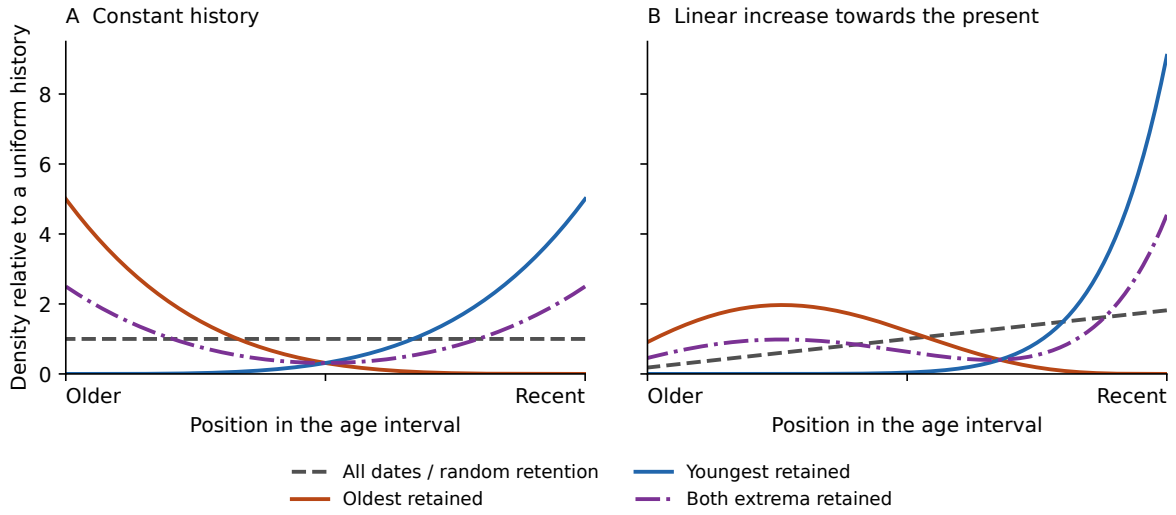


Figure 1: How retaining extremes changes an age distribution. Each hypothetical site has five independent, exactly observed ages from a constant history (A) or a linear history with ten-fold greater density at the recent end (B). Curves are analytic densities from the order-statistic equations in Supplementary Section S2.6, normalised to unit integral over the age interval. Random retention has the same marginal density as the generating history. Keeping both extrema combines their two marginal densities equally. These curves illustrate selection before calibration; they are not fitted histories or results from the calibrated simulation.

458 **Fig. 2.**

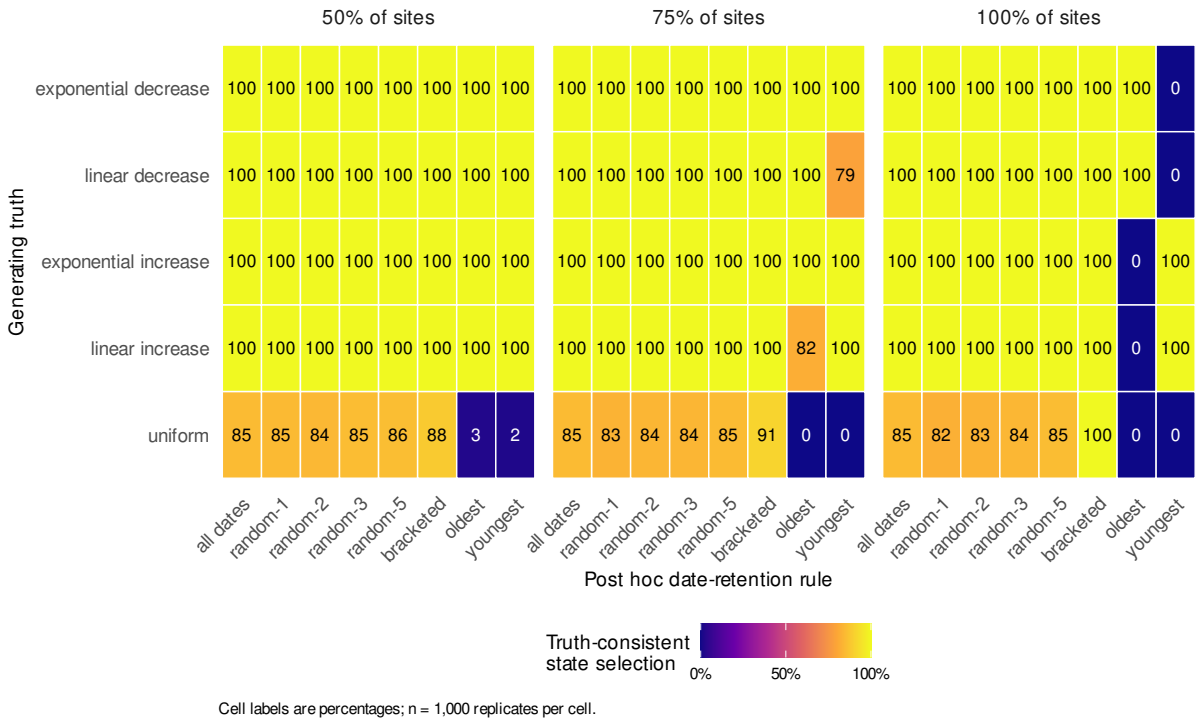
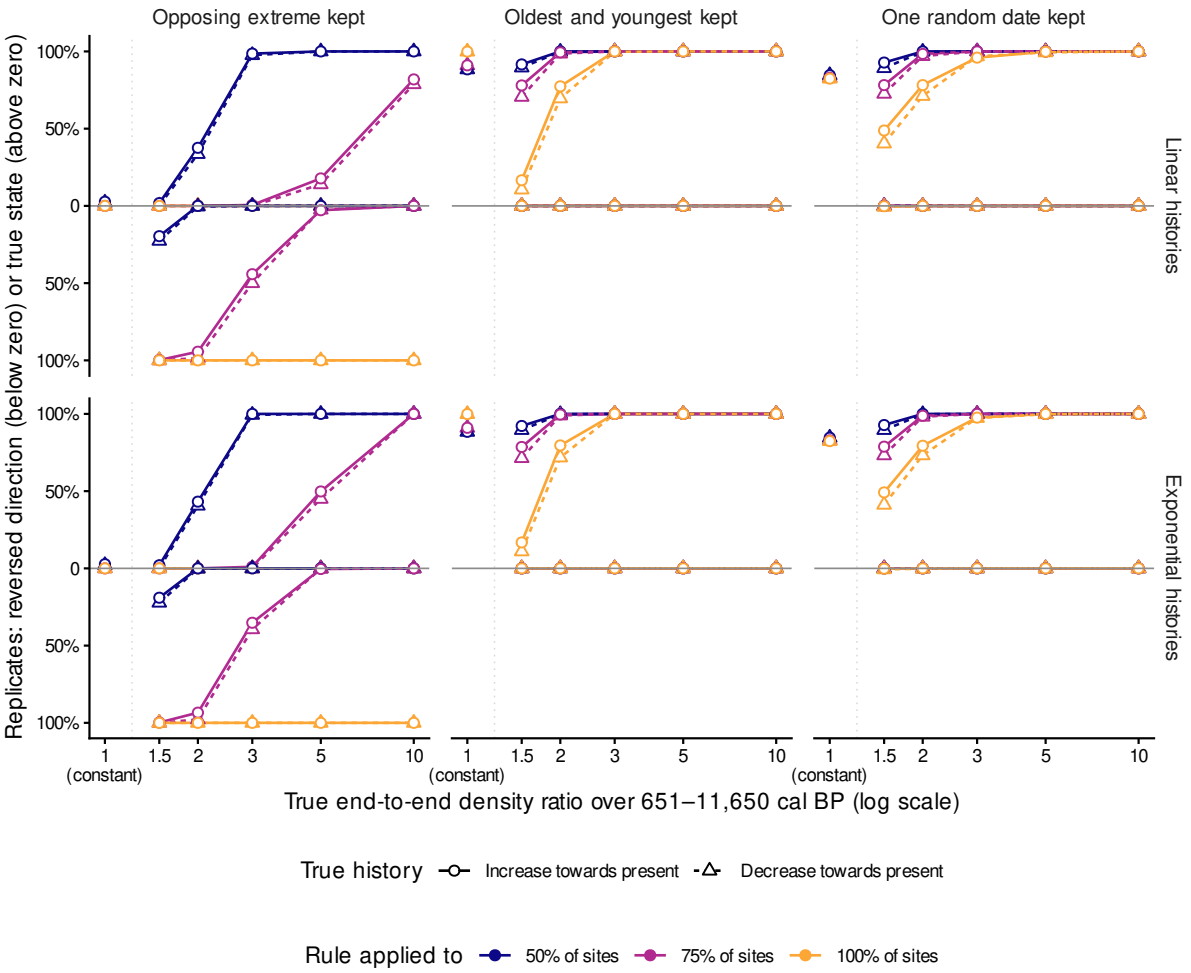


Figure 2: Correct-direction selection under post hoc date retention (labelled "truth-consistent state" in the figure). Rows are the ten-fold changing histories and constant reference; columns are retention rules; panels give the site-treatment proportion. For uniform truth, success means selecting uniform; for non-uniform truths, it means selecting the correct temporal direction, whether the selected family is linear or exponential. Cell labels are percentages from 1,000 replicates. Uniform selection is relative to the fitted candidates; no U-shaped alternative was fitted, so a high bracketed-uniform value does not establish shape recovery. Results use minimum AIC under the primary marginal likelihood fitted directly to calibrated posteriors; bracketed cells treat paired within-site extrema under working independence.

459 **Fig. 3.**



Opposing extreme: oldest date for increasing histories, youngest for decreasing. Above zero: AIC favours the correct direction, or uniform for the constant history (ratio 1, unconnected). Below zero: AIC favours the opposite direction. For ratios above 1, the remaining replicates favour uniform. Ratios 1.5–5: 500 replicates; ratios 1 and 10: 1,000.

Figure 3: Reversal of weaker trends under partial retention. For each true end-to-end density ratio (log scale), points above zero give the proportion of replicates in which AIC favoured the true state: the correct direction, or uniform for the constant history at ratio 1, shown unconnected. Points below zero give the proportion favouring the opposite direction; for changing histories, the remaining replicates favoured uniform. Columns show the opposing one-sided rule (oldest date for increasing histories, youngest for decreasing), both extrema, and one random date per site; colour gives the proportion of sites treated. Ratios 1.5–5 have 500 replicates per point and ratios 1 and 10 the accepted 1,000. Primary marginal likelihood; bracketed cells use working independence.

460 **Fig. 4.**

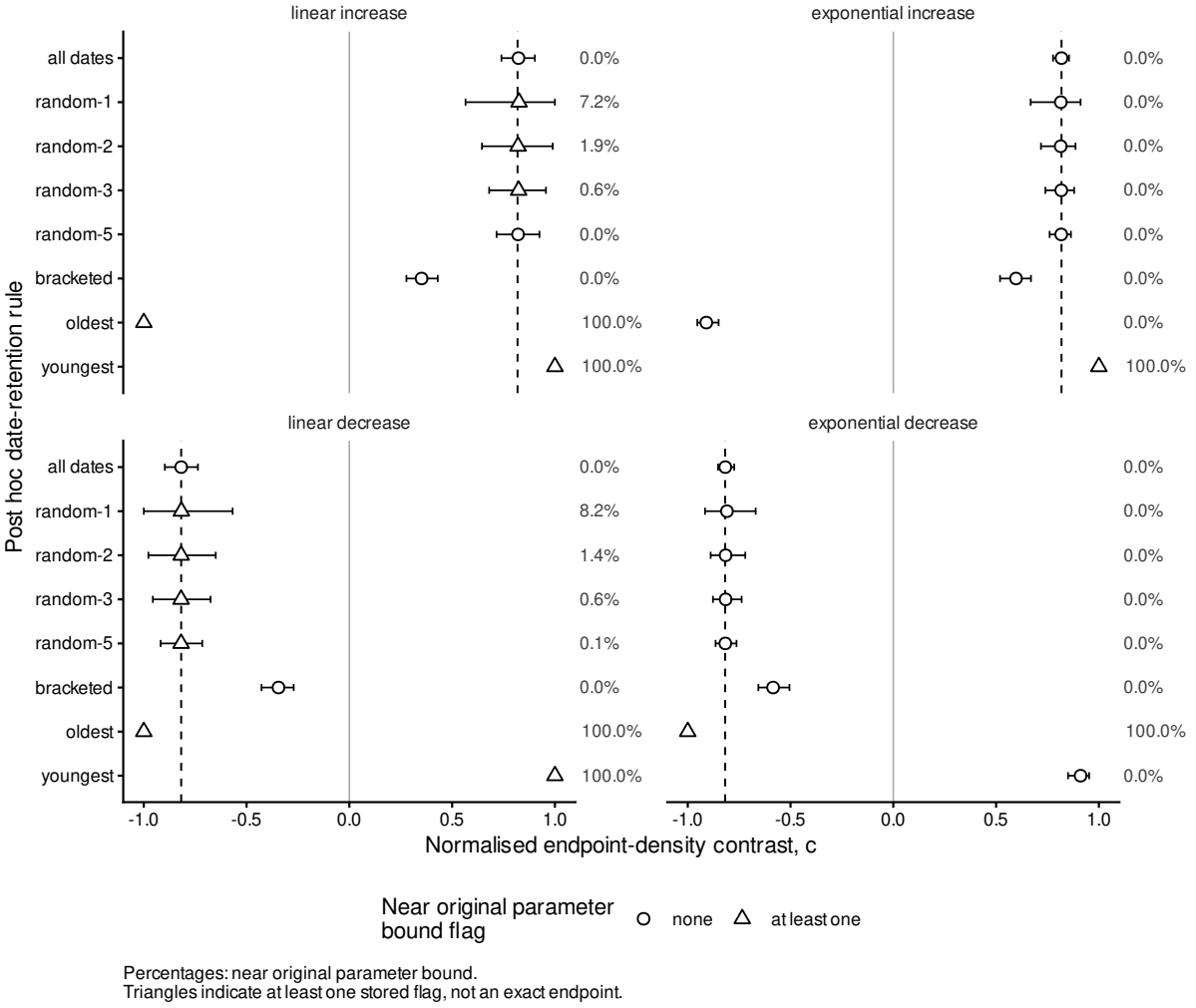


Figure 4: Fitted generating-family endpoint contrast for ten-fold changes when every site is subjected to the retention rule. The common bounded scale is $c = \{f(651) - f(11650)\}/\{f(651) + f(11650)\}$, with positive values indicating greater density at the recent end. Dashed lines mark the true $c = \pm 9/11$ and the grey line marks equal endpoint density. Points are means and bars are central 95% replicate ranges, calculated after transforming each of 1,000 fitted contrasts. Percentages give the fraction of fits near an original parameter bound, and triangles mark groups with at least one such fit. Linear bounds allow zero density at one endpoint ($c = \pm 1$, an infinite log ratio); the exponential rate cap gives a finite endpoint ratio. The bounded scale compresses large ratios. Estimates use the generating family even when AIC selected another family. Primary marginal likelihood; bracketed cells use working independence.

Fig. 5.

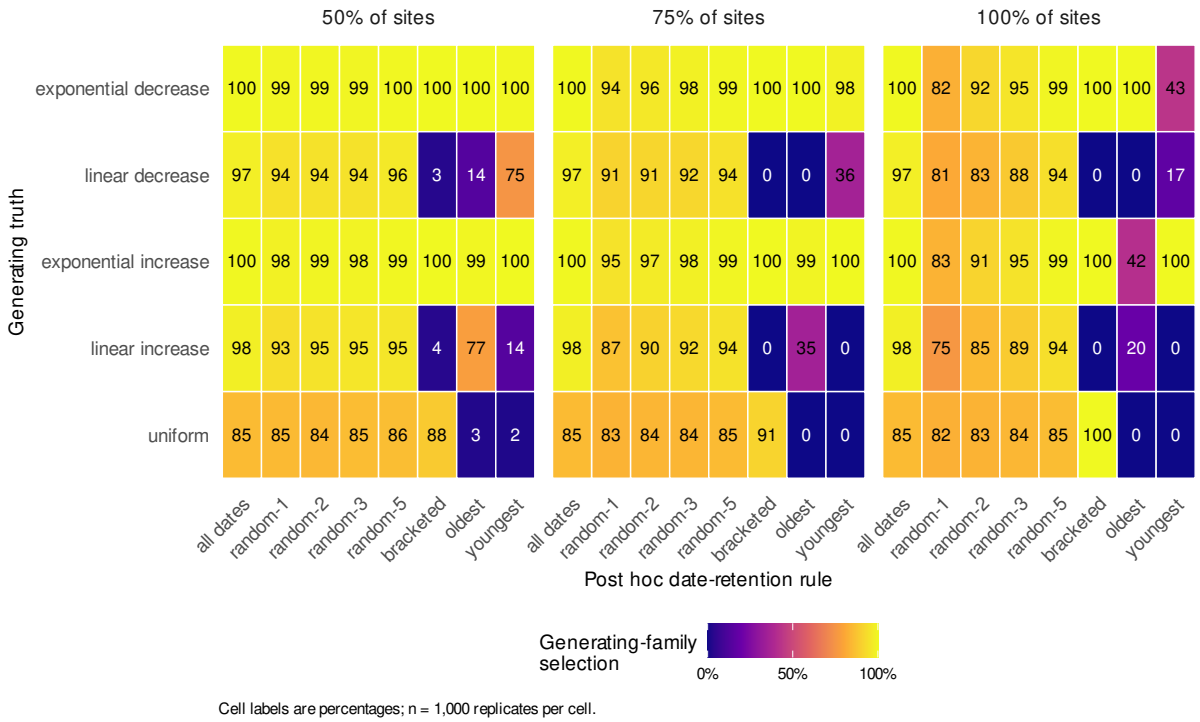


Figure 5: Generating-family selection under post hoc date retention. Rows are the ten-fold changing histories and constant reference; columns are retention rules; panels give the site-treatment proportion. Cell labels are the percentages of 1,000 replicates in which minimum AIC selected the generating family, irrespective of whether its fitted direction was correct. Comparing this figure with Figure 2 separates directional fidelity from linear-versus-exponential discrimination. Primary marginal likelihood; bracketed cells use working independence.

462 **Fig. 6.**

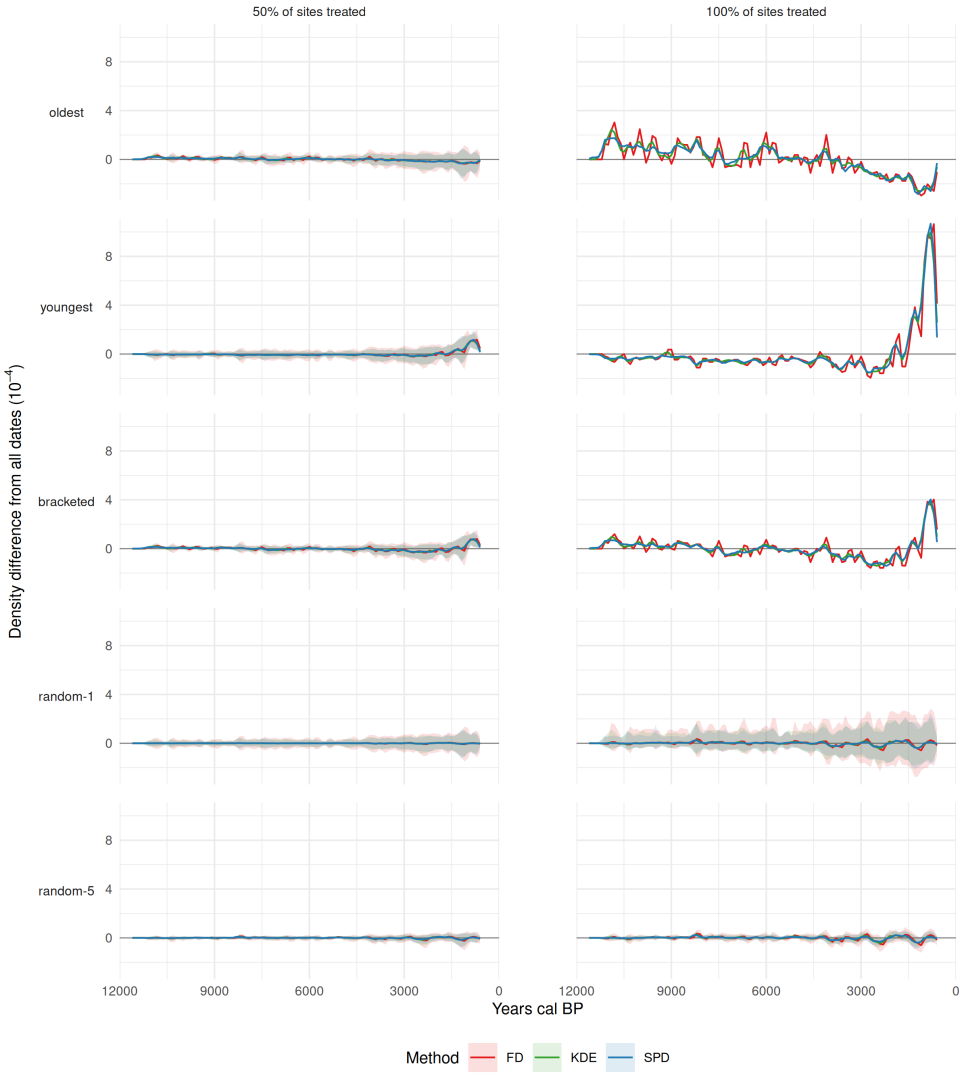


Figure 6: Change in empirical calendar-age density after within-site retention. Rows show oldest, youngest, bracketed (both extrema), random-1 and random-5 retention; columns show treatment of 50% or 100% of 68 sites. Curves subtract each method's all-date baseline from its retained unit-integral summary, so positive values mark calendar ages over-represented after retention. Lines are means and shading spans pointwise 2.5th–97.5th percentiles across 1,000 conditional realisations; the deterministic all-site extrema rules have no envelope. Envelopes describe retention variation, not uncertainty over sites or population history. SPD, FD and KDE receive identical dates; the shared vertical scale compresses smaller random-retention changes. At 100% treatment, oldest, youngest and random-1 each retain 68 dates, bracketed retains 136 and random-5 retains 340.